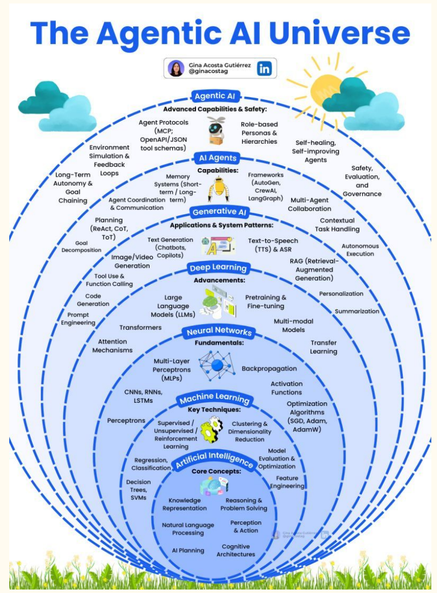
LLM Fine-Tuning: LoRA and QLoRA

By: Uday Goel, RRHS Capstone, Grade 12

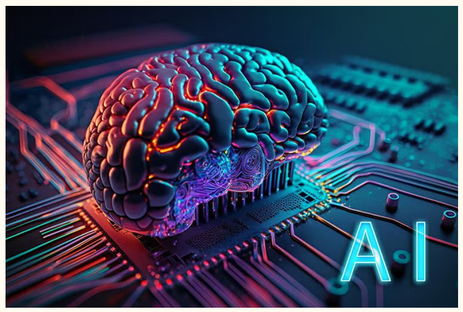
Essential Questions:

- How can LLM's be adapted more efficiently?
- Why is AI development focused within the top companies?
- What changes do LoRA and QLoRA bring to the world of AI?



The image to the write shows us a food chain type view of how AI works, from starting with some of the core terms for AI, then moving to machine learning and neural networks and so on. The main goal of the image is not just about different types of models but it's a connected ecosystem going towards Agentic AI at the top of the ecosystem. This shows us that the core foundational layers and going towards creating different agentic AI agents such which helps create more effective systems.

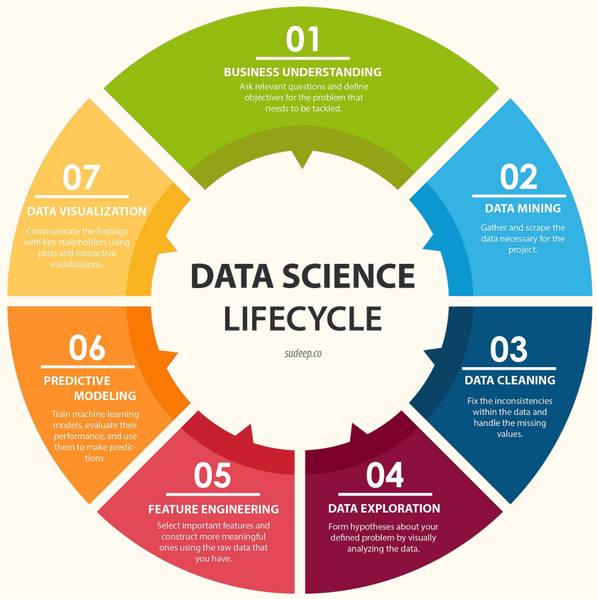
Background of AI Development



- 1763: Thomas Bayes, a English mathematician introduced the Bayes Theorem in statistics, useful in AI to handle uncertainty.
- 1950: Alan Turing, a British mathematician published a research paper called Computer Machinery and Intelligence, launched the field of AI.
- 1956: A group of scientists across the US met at Dartmouth for the Summer Research Project on AI, lead to the creation of the word: artificial intelligence.
- 1997: IBM's Deep Blue computer system defeated the world chess champion, Garry Kasparov.
- 2016: Google's DeepMind AI platform, AlphaGo defeated Go master Lee Sedol in a five-game challenge match in South Korea.
- 2020: OpenAI released GPT-3, the first version of ChatGPT, trained on over 175 billion parameters.
- 2023: QLoRA was introduced a fine-tuning technique to save money and storage.

What is Data Science

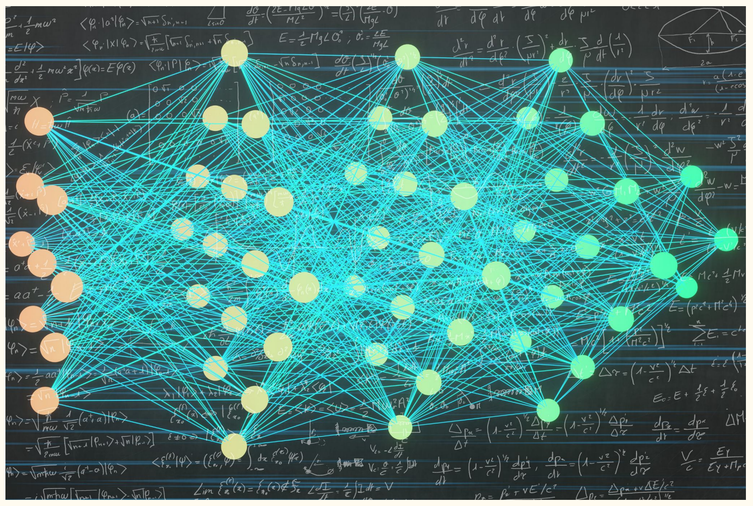
- Data science is a field of study which combines mathematics, statistics and computer science to be able to extract insights from any data which is presented.
- Data science also uses various ML and AI methods to understand the data better.
- Data scientists use a framework called the data science lifecycle which is a step by step process which shows you how to approach a new project or task you are trying to complete for data science.
- Along with the life-cycle which tells you how to approach a new project, their are also four pillars of data science which include:
- Mathematics
- Statistics
- Programming
- Communication

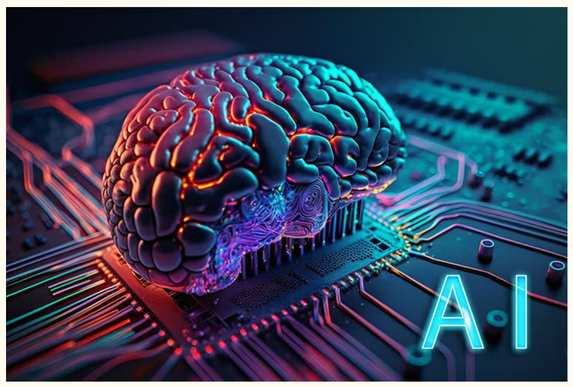


Data Science v.s Machine Learning v.s AI

1. Data Science:
- Uses stats, coding and math to understand data
2. Machine Learning:
- Algorithms which learn patterns on data
3. Artificial Intelligence:
- Simulate human-like intelligence through LLM's





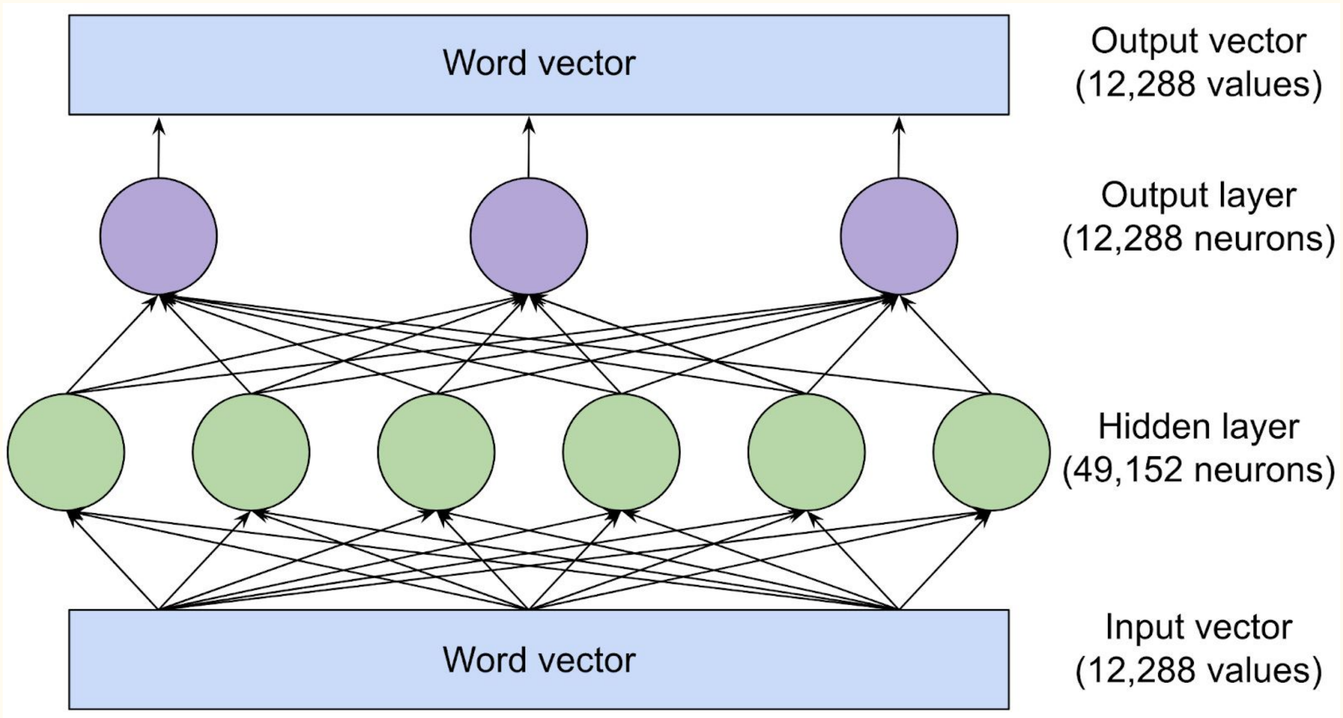


What are Large Language Models

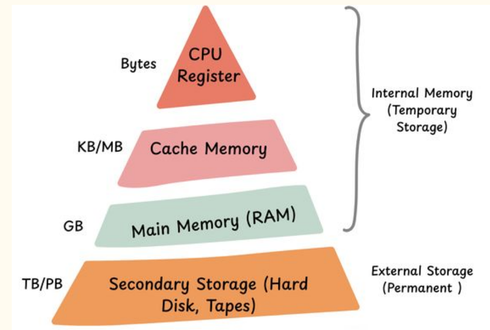
- Large language models are models which replicate many AI and ML systems as they are trained on large amounts of data, they are considered subsets of AI and ML.
- They are able to understand and generate human language which makes them useful for a variety of tasks.
- LLM's are important because they are able to connect human language and machine systems into a single technology.

Examples:

- ChatGPT - GPT-3
- Claude - Opus
- Gemini - Pro



The Problem with LLMs

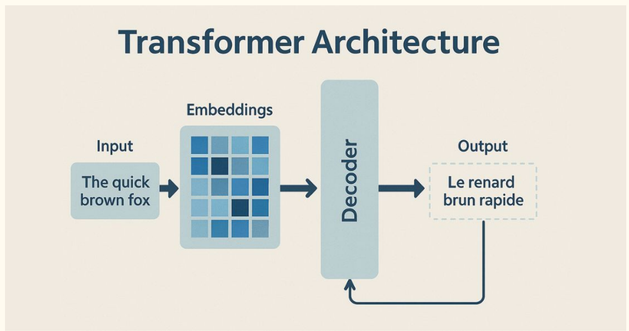


- The main problem surrounding large language models is large cost of training, storage and memory.
- Training GPT-3, OpenAI's first version of ChatGPT was estimated to cost between 4 and 12 million dollars, which is the type of money only large tech companies like Google and Apple have.
- Storage of LLMs is also another large problem as many LLMs have large hardware requirements with GPT-3 having close to 175 billion parameters which equals 350 GB of storage on a regular computer.
- Gemini has somewhere between 1.2 to 1.56 trillion parameters which equals close to 3.6 terabytes of storage on a computer.
- GPT-3 needed roughly 5 NVIDIA A100 for OpenAI
- Claude needed roughly 47 NVIDIA H100 for it's Opus model

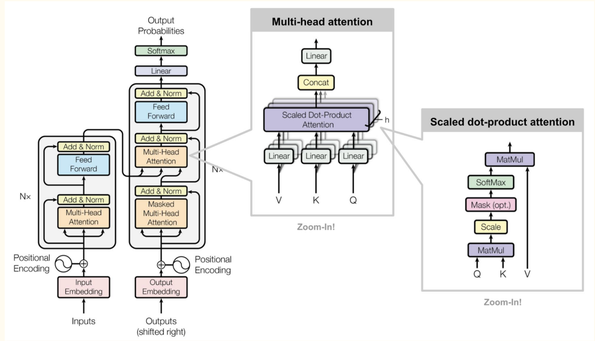
Transformer Architecture

Transformer Architecture is a type of design, which AI uses to process sequences of data, here is some vocab listed below:

- Token: Units of text in AI. This is usually a piece of word, 4 characters to 0.75 words. Handles any rare words or typos.
- Self-Attention: The brain of the transformer machine, analyzes other tokens which are part of the sequence to find out which ones are the most relevant.
- Query: Often considered as your search bar, what the token is looking for.
- Key: Seen as a video title or a tag, label which tells what this models offers.
- Value: Is the actual video, the content or meaning of the token.



How Transformers Work



- The best way to understand a transformer is to understand that it acts like a translator which is able to read an entire page at once and understand what is going on.
- A few key pieces which helps a transformer in doing it's job are shown below:

Input Tokens:

- Breaks sentence into smaller words, similar to vowels and constants to help pronounce complex words and sentences.

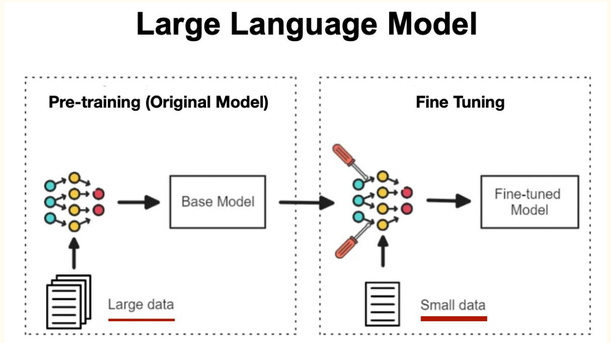
Attention Mechanism:

- Puts emphasis on the most important words in a sentence.

Output Predictions:

- Looks at the sentence as a whole and looks at the words coming up next.

What is Fine-Tuning



- Fine-tuning involves taking a pre-trained model like Gemini Pro or GPT-3 and giving it extra training on a specific dataset to refine it's skills or behavior.
- In much simpler terms, this is like teaching AI a specific new job.

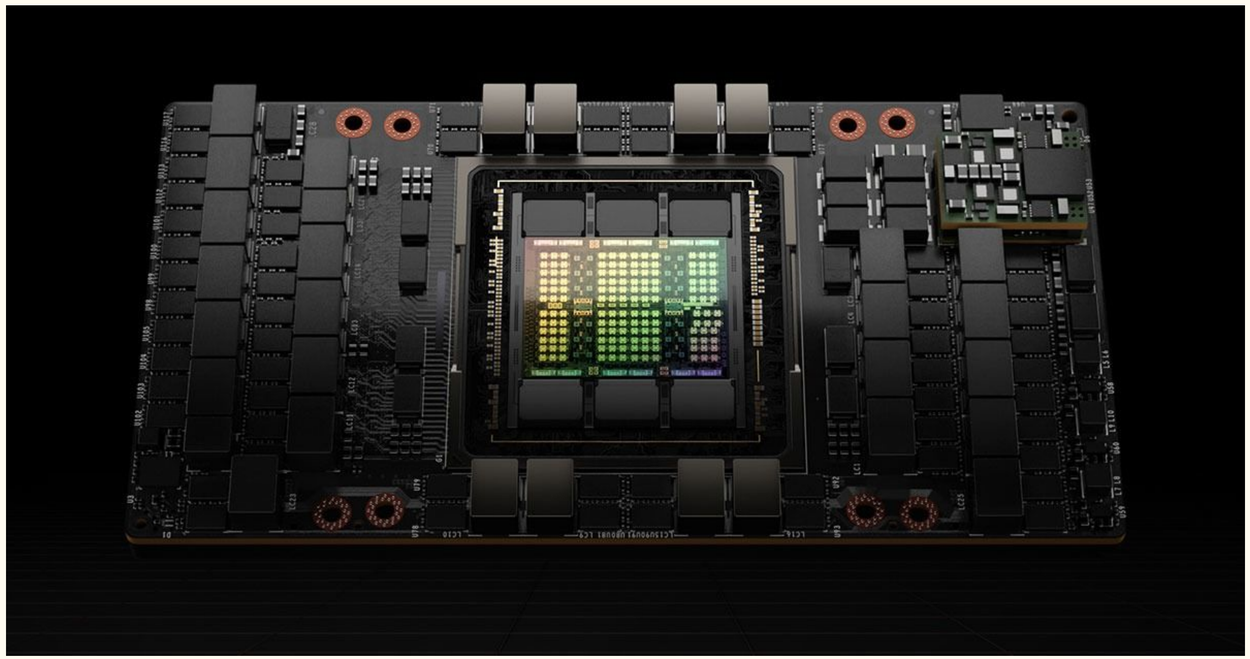
Medicine: Many base models are fine-tuned on a wide variety of medical images and lab results, this helps many doctors detect any early signs of a disease such as lung cancer or any heart failures, the accuracy levels are currently exceeding 90%.

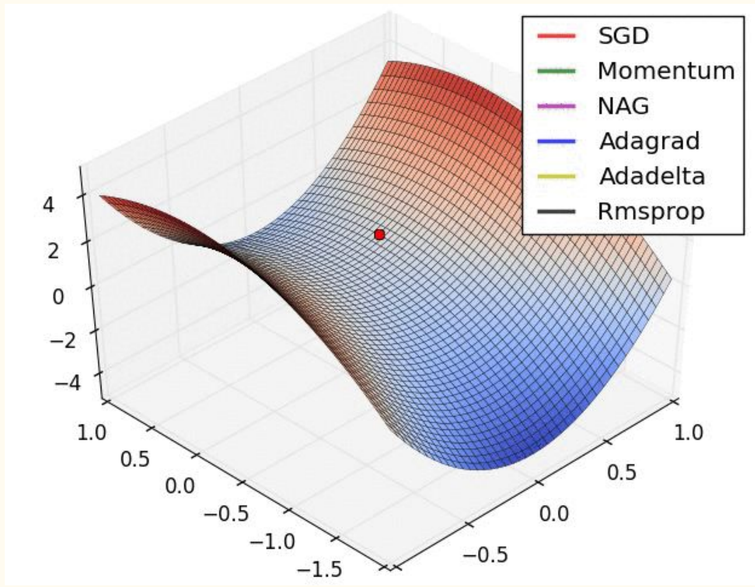
Law: A base model is fine-tuned on a large amount of legal contracts and fillings from various clauses in our legal system. The system is able to identify any missing terms or flag a certain clause while saving a fraction of time in comparison to a lawyer.

Education: Fine-tuned on a specific textbook along with teacher plans to help provide explanations which align with a school's teaching methods.

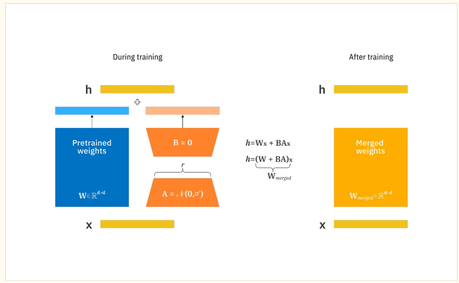
Why is Fine-Tuning So Expensive

- For fine-tuning a 65B parameter model, it requires a large amount which is 780 GB of VRAM, this is over 6 times more memory than inference.
- This is also because of the need to store 130 GB gradients and also a large amount of optimizer states and model weights.
- Due to the large amount of memory required, this needs high-cost infrastructure such as NVIDIA H100 GPUs which cost tens of thousands of dollars to buy.

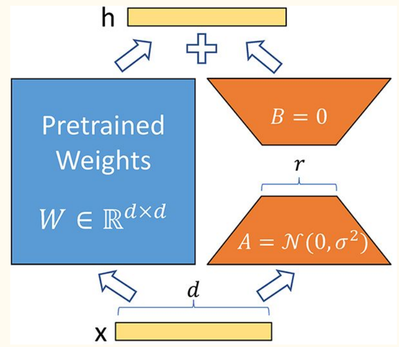




LoRA



- LoRA stands for Low-Rank Adaption, LoRA in much simpler terms is a method used to fine-tune a large AI model by only changing a small number of training parameters.
- LoRA reduces the number of trainable parameters for downstream tasks. The number of trainable parameters are reduced by 1000 times and the GPU memory requirement is decreased by 3 times.
- LoRA overall performs better on fine-tuning model quality despite having less parameters and having a higher training output.
- Key Idea: LoRA freezes the original model to only train smaller adapter matrices
- LoRA freezes the pre-trained model weights and put trainable rank decomposition matrices into each layer of transformer architecture.

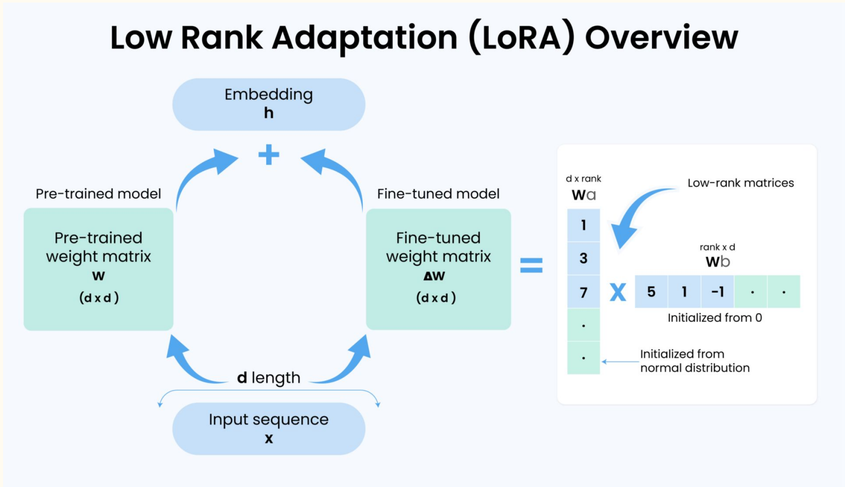


How LoRA Works

$$W_0x + \Delta W x = W_0x + BAx$$

- LoRA allows for training a dense layer in a neural network by using rank decomposition which changes during adaptation which keeps the pre-trained weights frozen, making the system both storage and compute efficient.
- Key Point: LoRA helps solve the problem for the large cost of computing and storage which is required to fine-tune AI models. This is done through freezing the base model and only training smaller adapter layers. This not only reduce memory usage but it also allows for fast task-switching.
- The main equation is shown in the top corner here is a layout of what each of the variables mean:
- X - Input Vector
- deltaW - Change applied to weights
- A - Low-Rank Matrix 1, uses Gaussian initialization
- B - Low-Rank Matrix 2, initialized to zero

LoRA Results



Some of LoRA's advantages include:

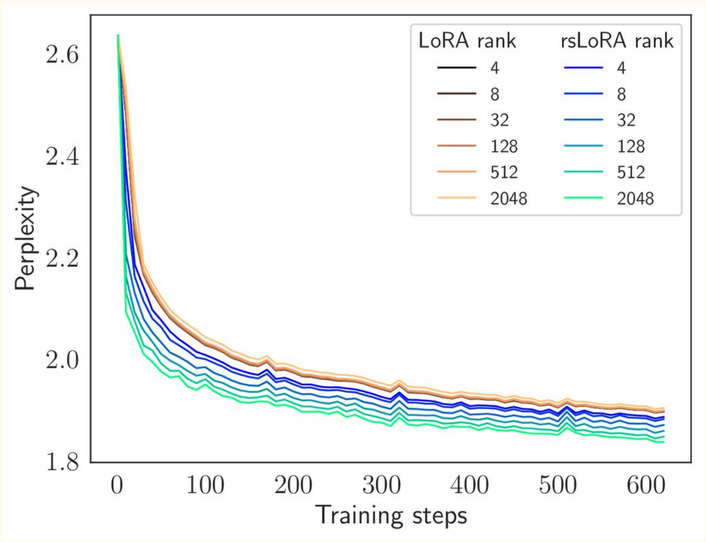
- Fast task-switching
- Training is more efficient, lower hardware barrier
- Can merge trainable matrices with frozen weights when deployed.
- LoRA is orthogonal, completely independent
- 99 to 99.9% fewer trainable parameters
- 3 times lower memory usage
- Similar benchmark performance
- Performed on par or better than full fine-tuning across all the models which were tested.

LoRA HyperParameters

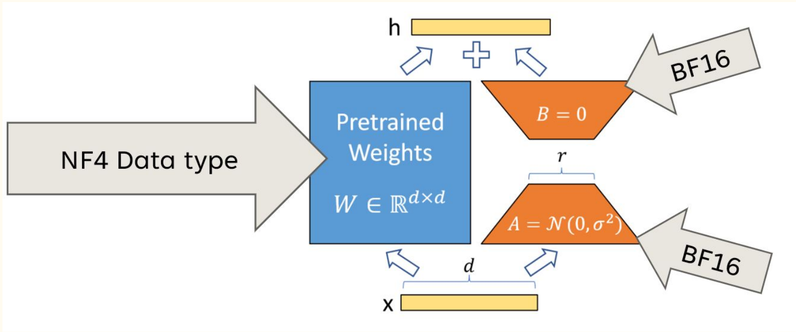
Some of the hyperparameters in LoRA include the rank (r) and alpha variables (α).

Rank (r) : Controls how many dimensions the adapter matrices use. Lower rank means fewer parameters, faster training, less capacity.

Alpha (α) : Controls the magnitude of the adapter's influence on the model output. Often set equal to r. Think of rank as the resolution (how much detail you store) and alpha as the volume (how loudly the adapter speaks relative to the base model).



QLoRA



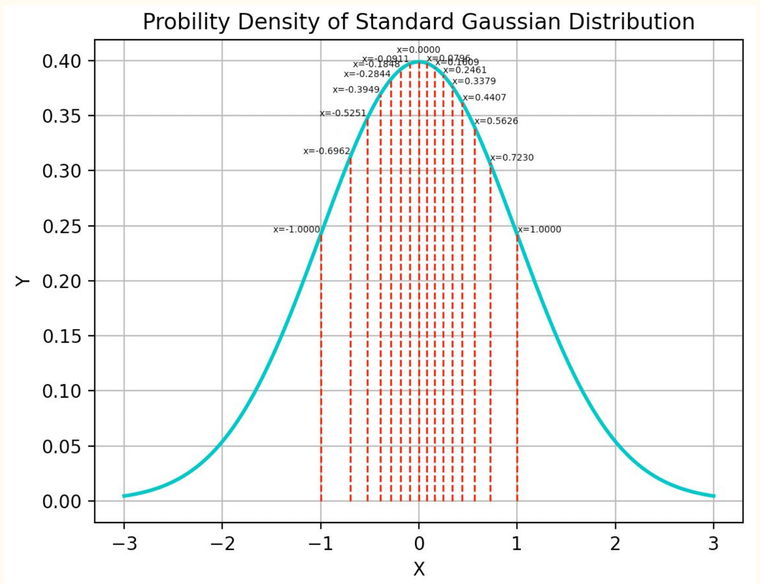
- QLoRA stands for Quantized Low-Rank Adaption and it's a fine-tuning approach that reduces memory usage, fine-tunes a 65B parameter model on a single 48 GB GPU while preserving the full task performance.
- The best model family is Guanaco which outperforms all previous openly released models, reaches close to 99% of the performance of the GPT while only needing 24 hours of fine-tuning.
- QLoRA has three innovations which helps save memory:

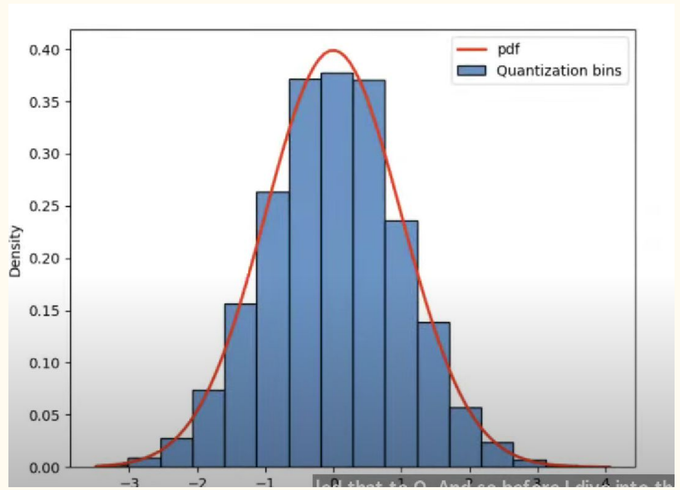
1. 4-bit Normal Float (NF4)
2. Double Quantization
3. Paged Optimizers

Main Goal: To democratize the fine-tuning of AI models. Reduces the memory and resources needed, allows developers to fine-tune large LLMs on consumer-grade hardware.

Normal 4-Bit (NF4) Quantization

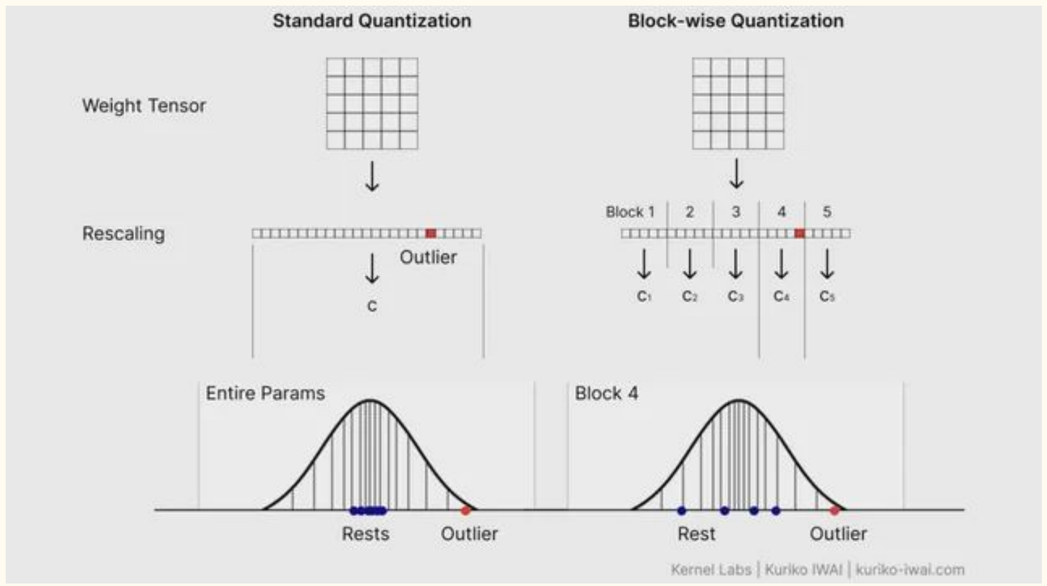
- A 4-bit data type introduced in QLoRA that is used for normally distributed weights. The standard 4-bit quantization spaces its 16 levels evenly across the data range.
- NF4 instead places levels at the quantiles of a standard normal distribution, so each level covers an equal expected mass.
- Since neural network weights follow approximately normal distributions, NF4 incurs less rounding error than standard 4-bit.
- This data type is for normally distributed data, much better results than 4-bit Integer and floats.





Double Quantization

- Applying quantization to the quantization constants themselves.
- Block-wise quantization requires one scaling constant per block of 64 parameters, stored at 32-bit. For a 65B model, these constants occupy ~4GB.
- Double quantization re-quantizes them to 8-bit with block size 256, reducing per-parameter overhead from 0.5 bits to 0.127 bits and saving ~3GB.
- Quantized the quantization constants, reduces the average memory footprint.



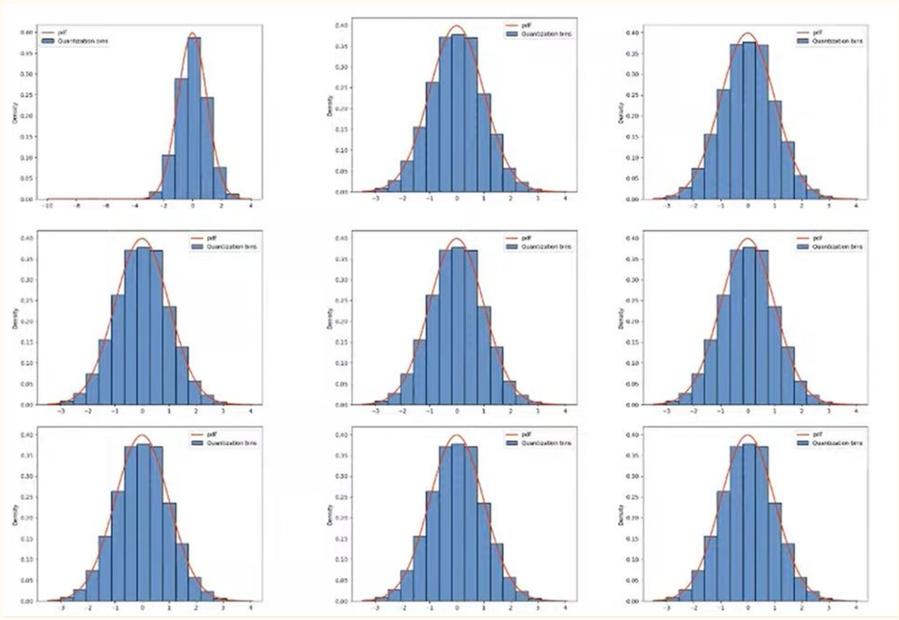
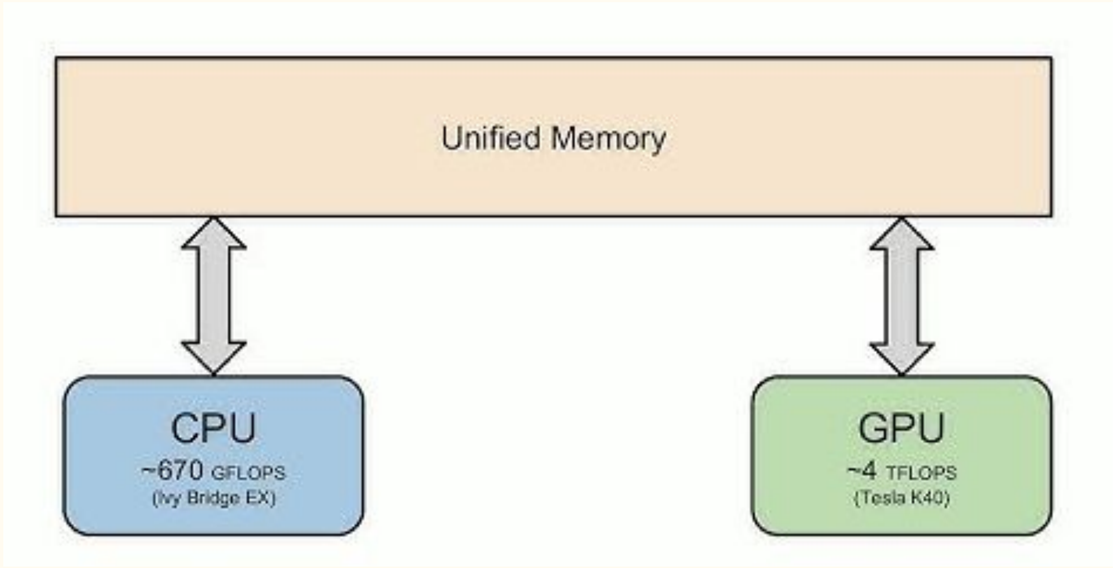


Image source: *Democratizing Foundation Models via k-bit Quantization by Tim Dettmers*

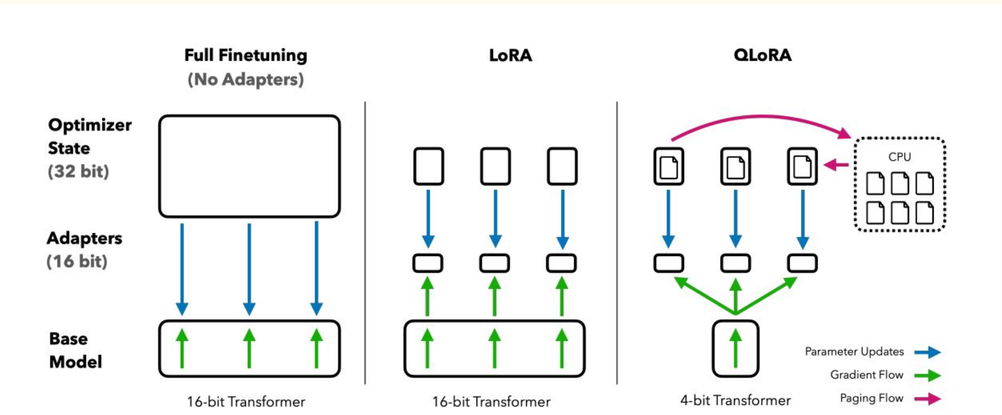
Paged Optimizers

- A mechanism using NVIDIA's unified memory (a single virtual address space shared by CPU and GPU) to automatically migrate optimizer states from GPU to CPU RAM when the GPU runs out of memory during a long-sequence mini-batch.
- Then retrieve them when needed. Prevents training crashes without changing the optimizer algorithm. Analogous to OS virtual memory paging.
- Used primarily for memory spikes, usually from sharp increase in amount of RAM.
- NVIDIA Unified Memory: Memory management feature that creates a single, virtual address space for CPU and GPU.



Guanaco Results

- Guanaco which is a series of AI chatbots including: LLAMA, OSSTI DataSet, it was a conversation designed to train AI models.
- Training took under 24 hours on a single 48 GB A100 GPU, cost around 300 dollars.
- From Guanaco model testing, it was able to find that for question like the capital of Zambia, the models answered correctly. It was also good with other normal activities.
- It will not answer common questions like how are you, cannot be trusted with secrets. Strong their of mind, answers philosophical questions very well.
- Achieved 99.3% of ChatGPT's performance.



LoRA vs QLoRA vs Full Fine-Tuning

Full-Fine Tuning:

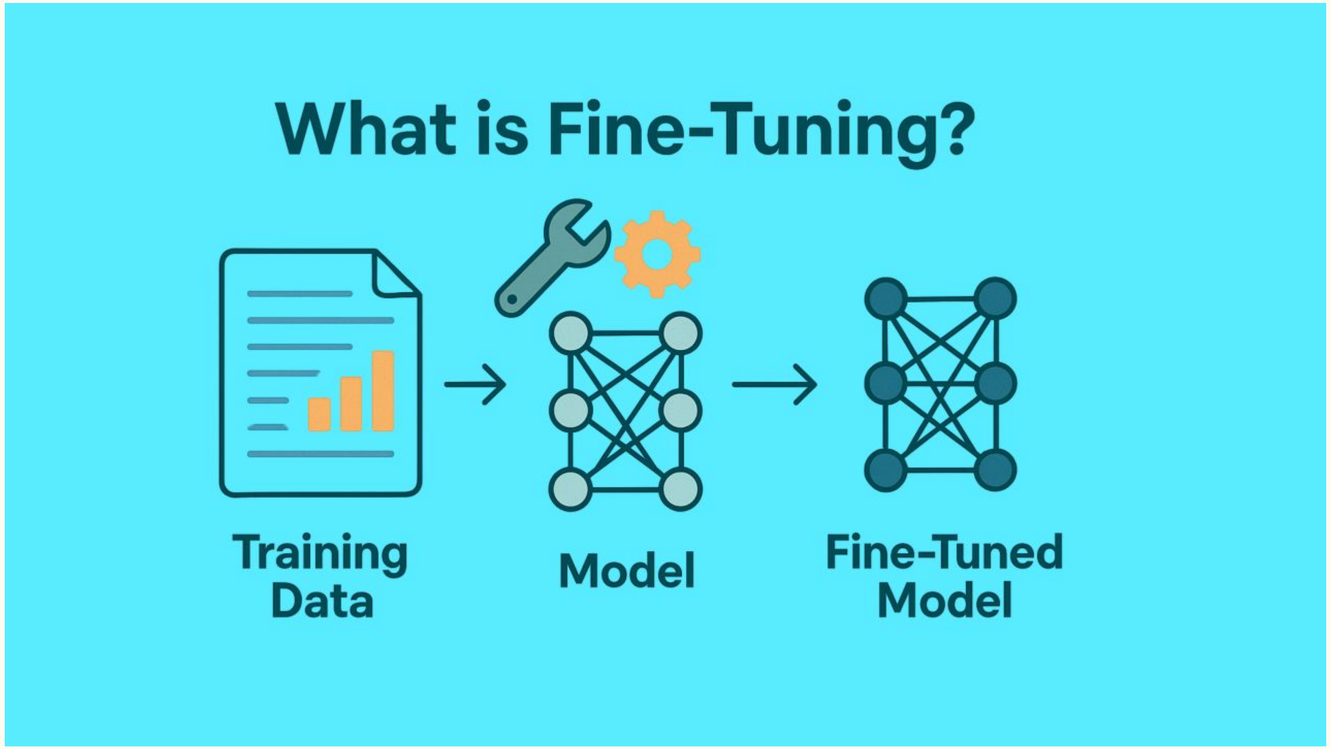
- Cost: High
- VRAM: 780 GB
- Performance: Baseline

LoRA:

- Cost: Medium
- VRAM: 131 GB
- Performance: Similar

QLoRA:

- Cost: Low
- VRAM: 33 GB
- Performance: Similar



Limitations

- LoRA's low-rank constraint underperforms on tasks requiring new knowledge as opposed to tasks requiring stylistic or behavioral adaptation. LoRA's constraint acts as a regularizer that preserves pre-training knowledge at the cost of task-specific adaptation capacity.
- QLoRA's 4-bit NF4 quantization introduces quantization noise that can affect model performance on precision-sensitive tasks, particularly those involving fine-grained numerical reasoning or highly specialized vocabulary.
- The $300 fine-tuning cost is not the deployment cost. Serving a 65B model at production inference scale requires persistent GPU infrastructure GPU hours, latency optimization, and reliability engineering that can exceed the one-time fine-tuning expense over time.
- LoRA and QLoRA depend entirely on open-weight foundation models. LLaMA-65B cost Meta an estimated $5 million or more to pre-train.

Future of Fine-Tuning

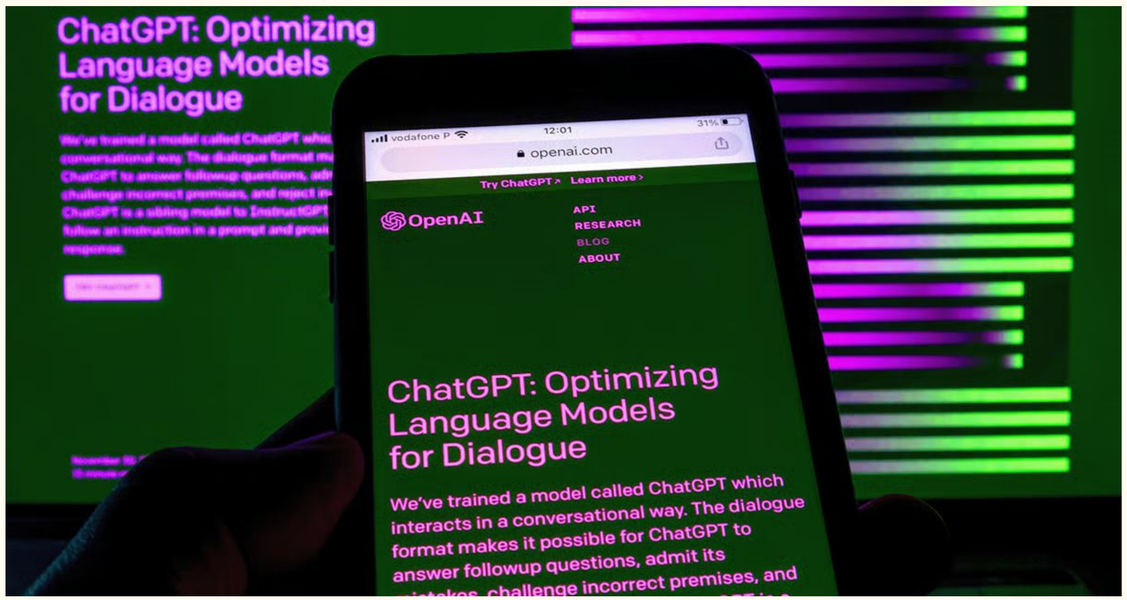
DoRA: Addresses LoRA's expressiveness limit by decomposing each weight matrix into separate magnitude and direction components, applying low-rank adaptation to the directional component only. Full fine-tuning modifies both magnitude and direction; LoRA modifies only a low-rank blend of both. DoRA's decomposition consistently outperforms LoRA across reasoning, visual instruction tuning, and various other benchmarks, it often approaches full fine-tuning performance more closely.

AdaLoRA: Solves the uniform rank allocation problem in standard LoRA, where the same rank r is applied to all adapted weight matrices regardless of how important each matrix is. AdaLoRA initializes adapters with higher rank and cuts out singular values during training based on computed importance scores, dynamically allocating the rank budget toward matrices that benefit most. Leads to better performance than uniform-rank LoRA at the same total budget.

Why I Chose This

I chose this topic because:

- Had a strong background and interest in ML and Data Science
- Interested in advanced AI technologies
- Wanted to know how ChatGPT and LLM's Work
- Wanted to know more about how smaller researchers competed with OpenAI and Google





What Did I Do

Week 1:

- Studied AI history
- Learned about transformer architecture
- Learned about how LLM's are trained

Week 2:

- Read LoRA research paper
- Read QLoRA research paper
- Read other sources regarding AI and ML

Week 3:

- Combined all research into a presentation

What Did You Learn Pt.1

- The biggest thing I learned is that LoRA and QLoRA are changing the world of AI. I also learned how LoRA is able to fine-tune models with its main equation involving frozen weights and two matrices labeled A and B.
- LoRA is able to achieve significant improvements when it comes to fine-tuning LLM's by freezing the pre-trained weights and inputting two trainable matrices named A and B. This is important as it changes the learning process from the pre-trained weights to a compact side of data.
- This is what allows a lot of developers to swap out the A and B matrices with a shared base model instead of creating a new model for every task.
- This as a whole helps solve the massive storage problem which was previously faced when it comes to fine-tuning a LLM as fine-tuning an LLM required massive data centers.

What Did You Learn Pt.2

- When it comes to QLoRA, there are three innovations which are changing how AI models are fine-tuned: The first is a 4-bit Normal float which helps maintain the precision and accuracy of a 16-bit model using only 4-bits of storage for each parameter.
- The second is Double Quantization which ends up saving up to 3 GB of VRAM on a 65B model which is a significant amount of memory and storage. The final innovation as part of QLoRA is paged optimizers which helps prevent out of memory errors which help prevent any crashes during memory spikes.
- Each innovation as a part of QLoRA helps in a different way as the normal 4-bit helps maintain high accuracy for LLMs, the double quantization helps save memory for fine-tuning LLMs and paged optimizers help prevent any crashes to the model during memory spikes which is a sudden increase in RAM usage.
- Along with learning about QLoRA and LoRA I also learned about a wide variety of tests which many AI companies such as OpenAI, Antropic and Google use to train their models on, this was really interesting as it shows you how a lot of these complex AI systems are built and how much time and effort goes into creating them.

Why This Matters

- This problem matters because with methods like QLoRA and LoRA, it allows every company, organization or university in the tech world to have a level and equal playing field.
- With many researchers at universities in Egypt, Kenya or Brazil being to finally have access to top tier technology allowing everyone to have the power to fine-tuning a LLM instead of just a select few people being able to fine tune a LLM.
- This is especially important because it democratizes AI, instead of paying a large amount of money to train a model, anybody can now adopt a foundation model to a specific domain.
- Foundation models include GPT 5.5, Claude Opus 4.8 and Gemini 3.5.
- Many AI models may not have knowledge in specific field like medicine or law, by fine-tuning the model you can teach the model your exact terminology and workflow.

How Does This Connect to My Future

- I'm an incoming freshman at the University of Pittsburgh in the fall and I am looking to pursue a double major in data science & physics and quantum computing.
- This project helped by giving me a glimpse of what research looks like, even though I was able to participate in data science research last summer with a professor from the University of Michigan, this was my first time conducting research on AI methods and understanding what fine-tuning means along with the various parameters which are needed to train an AI model.
- All of the technical content is directly relevant as LoRA, QLoRA, and transformer architecture are all foundational topics in modern machine learning, and understanding them now puts me ahead of most incoming freshmen.
- These are skills that transfer to every research project, every analysis, and every claim I will make as a data scientist. After graduating college, I want to work within either quantum computing or AI. This includes companies like IBM, Google, NVIDIA, Apple or Antropic.

Thank You - Questions



Please Ask Any Questions

Appendix

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